

Date: 23/12/2024

Course: B. Pharmacy

Subject Name: Biochemistry

Max Marks: 75

Semester: II

Subject Code: BP203T

Duration: 3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams/figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) Define carbohydrate with suitable example.
- ii) Differentiate between DNA and RNA.
- iii) State the coenzyme required for a conversion of pyruvic acid to acetyl co-A.
- iv) Define Protein. Classify them with examples.
- v) Give classification of enzyme with suitable example.
- vi) What is high rich compound? Give two examples.
- vii) Draw the structure of glucose.
- viii) What is hyperbilirubinemia?
- xi) State the enzymes present in DNA replication.
- x) Define rancidity with suitable example.

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- i) Explain the biosynthesis of 5-HT and melatonin.
- ii) Describe the reaction of urea cycle and mention in disorder.
- iii) Describe biosynthesis of Purine and Pyrimidine nucleotides.

Q.3. Short Answers (Answer 7 out of 9)

(7x5) = 35

- i) Explain in detail bile acid synthesis.
- ii) Describe HMP shunt pathway and give its regulation with significance.
- iii) Explain the disorder of lipid metabolism.
- iv) Explain about electron transport chain.
- v) Explain the synthesis and significance of adrenaline.
- vi) Explain the different type of RNA.
- vii) Explain genetic code gene its silent feature.
- viii) What are the substrate level phosphorylation and oxalate phosphorylation.
- xi) Define co-enzyme. Give the structure and biological role of NAD^+ and FAD^+ .